

Note 3 Solutions

Problem 1: Sum of Squares

Claim: For all $n \geq 1$,

$$\sum_{i=1}^n i^2 = \frac{1}{6}n(n+1)(2n+1). \quad (1)$$

Proof. We proceed by induction on n .

Base Case ($n = 1$):

$$\sum_{i=1}^1 i^2 = 1 = \frac{1}{6} \cdot 1 \cdot 2 \cdot 3 = 1. \checkmark$$

Inductive Hypothesis: Assume (1) holds for some $k \geq 1$.

Inductive Step: We prove (1) for $n = k + 1$.

$$\begin{aligned} \sum_{i=1}^{k+1} i^2 &= \sum_{i=1}^k i^2 + (k+1)^2 \\ &= \frac{1}{6}k(k+1)(2k+1) + (k+1)^2 \quad (\text{by inductive hypothesis}) \\ &= \frac{1}{6}(k+1)[k(2k+1) + 6(k+1)] \\ &= \frac{1}{6}(k+1)[2k^2 + 7k + 6] \\ &= \frac{1}{6}(k+1)(k+2)(2k+3), \end{aligned}$$

which is exactly (1) with $n = k + 1$. □

Problem 2: $3^n > 2^n$

Claim: For all $n \in \mathbb{N}$, $n \geq 1$, $3^n > 2^n$.

Proof. We proceed by induction on n .

Base Case ($n = 1$): $3^1 = 3 > 2 = 2^1$. $\checkmark$

Inductive Hypothesis: Assume $3^k > 2^k$ for some $k \geq 1$.

Inductive Step: We prove $3^{k+1} > 2^{k+1}$.

$$3^{k+1} = 3^k \cdot 3 > 2^k \cdot 3 > 2^k \cdot 2 = 2^{k+1},$$

where the first inequality uses the inductive hypothesis and the second uses $3 > 2$. □

Problem 3: Bernoulli's Inequality

Claim: For all $n \in \mathbb{N}$ and $x > -1$,

$$(1 + x)^n \geq 1 + nx. \tag{2}$$

Proof. We proceed by induction on n .

Base Case ($n = 0$): $(1 + x)^0 = 1 \geq 1 + 0 \cdot x = 1$. ✓

Base Case ($n = 1$): $(1 + x)^1 = 1 + x = 1 + 1 \cdot x$. ✓

Inductive Hypothesis: Assume $(1 + x)^k \geq 1 + kx$ for some $k \geq 1$.

Inductive Step: We prove (2) for $n = k + 1$. Since $x > -1$, we have $1 + x > 0$, so:

$$\begin{aligned} (1 + x)^{k+1} &= (1 + x)^k (1 + x) \\ &\geq (1 + kx)(1 + x) \quad (\text{by inductive hypothesis, since } 1 + x > 0) \\ &= 1 + kx + x + kx^2 \\ &= 1 + (k + 1)x + kx^2 \\ &\geq 1 + (k + 1)x, \end{aligned}$$

where the last inequality uses $kx^2 \geq 0$. □

Problem 4: $n! < n^n$

Claim: For all $n \in \mathbb{N}$, $n \geq 2$, $n! < n^n$.

Proof. We proceed by induction on n .

Base Case ($n = 2$): $2! = 2 < 4 = 2^2$. ✓

Inductive Hypothesis: Assume $k! < k^k$ for some $k \geq 2$.

Inductive Step: We prove $(k + 1)! < (k + 1)^{k+1}$.

$$\begin{aligned} (k + 1)! &= k! \cdot (k + 1) \\ &< k^k \cdot (k + 1) \quad (\text{by inductive hypothesis}) \\ &< (k + 1)^k \cdot (k + 1) \quad (\text{since } k < k + 1 \text{ implies } k^k < (k + 1)^k) \\ &= (k + 1)^{k+1}. \end{aligned}$$

□

Problem 5: Celebrity Problem

Definition: A *celebrity* at a party of n people is a person whom everyone knows, but who knows no one.

Algorithm

Input: n people $\{p_1, p_2, \dots, p_n\}$.

Output: The celebrity if one exists, or “no celebrity.”

1. **Elimination phase:** Maintain a single candidate, starting with p_1 . For each subsequent person p_i ($i = 2, \dots, n$), ask the current candidate if they know p_i :
 - If **yes**: the candidate cannot be a celebrity (celebrities know no one). Set p_i as the new candidate.
 - If **no**: p_i cannot be a celebrity (a celebrity must be known by everyone, but the candidate does not know p_i , so p_i is not universally known — wait, actually this eliminates p_i). Set the candidate unchanged.

After $n - 1$ questions, one candidate C remains.

2. **Verification phase:** We must verify C is actually a celebrity.
 - Check that C knows no one: ask C about each of the other $n - 1$ people. Note that the last elimination question already answered one of these (the question asked of C against the final eliminated person), so only $n - 2$ new questions are needed.
 - Check that everyone knows C : ask each of the other $n - 1$ people if they know C . This costs $n - 1$ questions.

If both checks pass, return C . Otherwise, return “no celebrity.”

Question Count

$$\underbrace{(n-1)}_{\text{elimination}} + \underbrace{(n-2)}_{\text{C knows no one}} + \underbrace{(n-1)}_{\text{everyone knows C}} = 3n - 4.$$

Proof of Correctness and Question Bound by Induction

Proof. We proceed by induction on n .

Base Case ($n = 2$): With two people, we ask one elimination question and one verification question, for a total of $2 = 3(2) - 4$ questions. The single remaining candidate is then correctly verified as a celebrity or not. ✓

Inductive Hypothesis: Assume the algorithm correctly identifies a celebrity (if one exists) among k people using at most $3k - 4$ questions.

Inductive Step: Given $k + 1$ people, ask one question to eliminate one person, reducing to k people. By the inductive hypothesis, the algorithm solves the k -person problem in at most $3k - 4$ questions. Then 2 additional verification questions are needed to check the eliminated person against the final candidate. The total is:

$$1 + (3k - 4) + 2 = 3k - 1 = 3(k + 1) - 4.$$

Correctness: Each elimination question removes exactly one non-celebrity: if A knows B , then A is not a celebrity; if A does not know B , then B is not a celebrity. Thus the final candidate is the only person who could be a celebrity. The verification phase then confirms whether they actually satisfy both celebrity conditions. $\square$